

Virgo Inference Summary

Notebook

virgo-inference(1).ipynb

Phase

Model Inference & Interactive Text Generation

Purpose

Loads the pretrained Virgo Base model and tokenizer, reconstructs the transformer architecture, and performs interactive text generation for inference and evaluation.

Objective

Provide an efficient inference pipeline for generating responses from Virgo Base using configurable decoding strategies.

Input Resources

virgo_base_final.pt checkpoint and virgo_tokenizer.json loaded from Kaggle datasets.

Model Loading

Reads model configuration from the checkpoint, reconstructs the decoder-only Transformer architecture, restores pretrained weights, moves the model to the available device (CPU/GPU), and switches to evaluation mode.

Architecture

Implements Rotary Position Embeddings (RoPE), Multi-Head Self-Attention, Position-wise Feed Forward Network, Transformer Blocks and the VirgoBase model.

Generation Pipeline

Prompt Encoding → Tokenization → Context Preparation → Autoregressive Token Generation → Decoding → Interactive Response Display.

Decoding Strategy

Supports temperature sampling, Top-k sampling, Top-p (nucleus) sampling, repetition penalty, configurable maximum generation length and BOS token handling for empty prompts.

Inference Configuration

Default parameters include max_new_tokens=256, temperature=0.8, top_k=50, top_p=0.95 and repetition_penalty=1.1.

Interactive Interface

Provides a command-line chat interface with a typewriter-style response animation and inference time measurement.

Outputs

Generated text responses, inference latency and decoded model predictions.

Engineering Highlights

Automatic device selection, checkpoint-based model reconstruction, efficient autoregressive decoding, configurable sampling strategies, tokenizer integration and interactive inference.

Summary

This notebook implements the complete Virgo inference pipeline, enabling efficient loading of pretrained weights and high-quality autoregressive text generation for interactive use.